Metric-Triggered WSD Decay for Language Model Pretraining

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***Abstract*—Warmup-Stable-Decay (WSD) learning rate schedules have emerged as a flexible alternative to cosine annealing for language model pretraining, decoupling the stable training phase from a short, sharp decay. Yet the key hyperparameters governing this decay—its timing, duration, and final learning rate ratio—remain largely empirical and are unlikely to generalize across models and datasets. In this work, we study the relationship between online training signals (loss variance and gradient signal-to-noise ratio) and the optimal decay trigger in WSD, using a LLaMA-style model trained on FineWeb. We reproduce scheduler comparison results from prior work, analyze how decay timing and magnitude affect final validation loss, and propose adaptive metric-based policies to automatically identify the "sweet spot" for initiating decay—the point at which the stable phase yields diminishing returns.**

I. Introduction

Cosine learning rate annealing [**?**] has long been the default schedule for pretraining large language models, but it imposes a hard token budget: the entire training duration must be fixed before training begins, and extending a run requires restarting from scratch. WSD schedules [**?**] address this limitation by separating training into three phases—warmup, a long stable phase at peak learning rate, and a short decay—making continued training and branching straightforward.

Despite this appeal, WSD introduces its own set of design choices. When should decay begin? How long should it last, and how deep should the learning rate drop? Existing work [**?**], [**?**] provides recommendations that are largely empirical and tuned for specific model and data scales. Finding the right timing—the *sweet spot*—is non-trivial: decaying too early forfeits further stable-phase progress, while decaying too late may leave potential gains on the table if the budget is fixed.

We define the sweet spot as the step at which the stable phase has converged to a plateau: further training at peak learning rate yields diminishing validation-loss improvements, and initiating decay will most efficiently compress remaining progress. Our work addresses three questions:

1) Can we reproduce the WSD vs. cosine comparison from prior work [**?**] on our model and dataset?
2) Do online training metrics correlate with validation loss under different decay timings and magnitudes, allowing us to predict the sweet spot?
3) Can we construct meaningful adaptive policies, based on these metrics, that match or improve upon fixed-schedule baselines?

II. Related Work

Learning rate scheduling is a critical and sensitive component of LLM pretraining [**?**], [**?**]. Cosine annealing has been the dominant choice [**?**], but it requires committing to a total token budget upfront, limiting training flexibility [**?**], [**?**].

To overcome this rigidity, minicpm introduced the WSD scheduler as part of the MiniCPM training framework. WSD keeps the learning rate at its peak value throughout a long stable phase and compresses the final descent into a short, steep decay, enabling seamless extension of training runs. wen2024river provided a theoretical grounding for WSD through the *river valley* loss landscape perspective: the stable phase navigates the model along a flat valley bottom, while decay rapidly descends to a sharp minimum. Their analysis, conducted under SGD dynamics, also introduced WSD-S—a smoothed variant of the decay curve—and empirically compared WSD-S against cosine annealing, demonstrating faster convergence per token. Further empirical evidence for WSD's efficiency has been reported in a number of subsequent pretraining studies.

All of this prior work treats the timing and magnitude of the decay as fixed, hand-tuned hyperparameters. Our study fills this gap by asking whether *online* training signals can replace these manual choices.

III. Materials and Methods

A. Datasets

We use FineWeb [**?**] (`sample-100BT`, 100B tokens) as our primary pretraining corpus, a large-scale, high-quality web crawl widely adopted in recent open LLM research. To test cross-dataset generalization of our policies, we also run selected experiments on C4 [**?**], another standard English pretraining benchmark. Both datasets offer sufficient scale for our computation budget while remaining representative of realistic pretraining conditions.

B. Model

We train a LLaMA-style [**?**] causal language model with 8 transformer layers, hidden dimension 512, intermediate size 2048, and 8 attention heads, yielding approximately 40M parameters. This size provides an acceptable trade-off between LLM representativeness and the constraint of running many ablation experiments.

We use the GPT-2 tokenizer (vocabulary size 50,257) and a context length of 1024 tokens. Optimization uses AdamW [**?**] ($\beta_1 = 0.9$, $\beta_2 = 0.95$, weight decay 0.1) with a peak learning rate of 6×10^{-4} and gradient clipping at norm

1.0. Following wen2024river, we adopt AdamW despite the WSD theory being developed under SGD, as AdamW is well established for LLM pretraining. Effective batch size is 524,288 tokens (batch 8, accumulation 64), giving a total token budget of 5.24B tokens over 10,000 steps. We use 5,000 validation samples sampled from a held-out stream of FineWeb, evaluated every 50 steps. Warmup lasts 100 steps.

C. Experimental Setup

All experiments share the base configuration above unless stated otherwise. We design four experiments to progressively build from reproduction to adaptive policies.

1) Exp. 1 — Scheduler Comparison: We reproduce the cosine vs. WSD-S comparison of wen2024river on our model. A single WSD-S run (decay starting at 90% of steps, lasting 10%, final LR ratio 0.1) is compared against a cosine run with identical peak LR and training budget. This follows the paper's recommended hyperparameters and confirms that our setup reaches the plateau phase before decay.

2) Exp. 2 — AdamW vs. Quasi-SGD Decay: The theoretical analysis of WSD in wen2024river assumes SGD. To better align with this theory during the decay phase, we shrink the AdamW momentum coefficients to $(\beta_1, \beta_2) \approx (0, 0)$ at the moment decay begins, effectively disabling first- and second-moment accumulation and recovering a quasi-SGD update rule (see Appendix **??** for the derivation). We compare three conditions: (a) WSD with standard AdamW, (b) cosine with standard AdamW, and (c) WSD with AdamW betas shrunk at decay onset.

3) Exp. 3 — Decay Timing and Amount Sweep: To map how the sweet spot depends on when and how strongly the decay is applied, we run a grid of WSD-S runs varying the decay start step (as a fraction of total training: 50%, 70%, 80%, 90%) and the final LR ratio ($r \in \{0.5, 0.3, 0.1, 0.05, 0.01\}$). For each configuration we record the final validation loss and the values of two online metrics at decay onset: *loss variance* (variance of training loss over a 50-step window) and *gradient SNR* (ratio of the mean gradient norm to its standard deviation across layers). The goal is to find reliable correlations between these pre-decay signals and post-decay performance.

4) Exp. 4 — Adaptive Decay Policies: Based on the correlations found in Exp. 3, we propose three adaptive policies that monitor training in real time and trigger decay automatically once the signal indicates the sweet spot has been reached:

- **Loss-variance policy**: triggers when the current `loss_variance` falls below the 3rd percentile of values observed in the preceding lookback window (10% of total steps).
- **Grad-SNR policy**: analogously triggers on the 3rd percentile of `grad_snr`.
- **Combined policy**: requires *both* conditions simultaneously.

All policies are gated: they cannot fire before 70% of total steps (7,000), to avoid premature decay. Decay length is fixed at 1,000 steps with final LR ratio 0.1. Percentile thresholds are computed from the lookback window at gate time (step 7,000).

[Fig. 1: Validation loss curves for cosine vs. WSD-S, Exp. 1 reproduction of wen2024river]

Fig. 1. Cosine vs. WSD-S validation loss on FineWeb (10k steps). Decay starts at step 9,000 (90%) for WSD-S.

[Fig. 2: WSD standard AdamW vs. WSD quasi-SGD betas vs. cosine, validation loss, Exp. 2]

Fig. 2. Effect of shrinking AdamW betas at decay onset to approximate SGD dynamics as assumed by the WSD theoretical analysis.

We compare each policy against three fixed baselines that start decay at 70%, 80%, and 90% of steps respectively—the last being the setting recommended by wen2024river.

IV. Results

Figures **??**–**??** summarize the results of each experiment in the order described above.

V. Discussion

VI. Summary

[Fig. 3: Final validation loss vs. decay start/amount, and metric correlations, Exp. 3]

Fig. 3. Heatmap of final validation loss across decay timing × final LR ratio, with correlation of pre-decay loss variance and grad SNR to final loss overlaid.

[Fig. 4: Adaptive policies vs. fixed baselines, validation loss, Exp. 4]

Fig. 4. Validation loss for adaptive policies (loss variance, grad SNR, combined) vs. fixed baselines at 70%, 80%, and 90% trigger.

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Appendix

The AdamW update at step t is:

$$m_t = \beta_1 m_{t-1} + (1-\beta_1)\, g_t, \tag{1}$$

$$v_t = \beta_2 v_{t-1} + (1-\beta_2)\, g_t^2, \tag{2}$$

$$\hat{m}_t = m_t/(1-\beta_1^t), \quad \hat{v}_t = v_t/(1-\beta_2^t), \tag{3}$$

$$\theta_t = \theta_{t-1} - \eta\, \hat{m}_t/(\sqrt{\hat{v}_t}+\varepsilon) - \eta\, \lambda\, \theta_{t-1}, \tag{4}$$

where g_t is the gradient, η the learning rate, and λ the weight decay coefficient. When $\beta_1 \to 0$, the first moment collapses to the instantaneous gradient: $m_t \approx g_t$, eliminating gradient averaging. When additionally $\beta_2 \to 0$, the second moment tracks $v_t \approx g_t^2$, so the adaptive scaling $\hat{m}_t/\sqrt{\hat{v}_t}$ approaches $\text{sign}(g_t)$—a per-element signed update reminiscent of SignSGD.

In practice, setting $(\beta_1, \beta_2) \leftarrow (0, 0)$ at the onset of the decay phase effectively disables the exponential moving averages built up during the stable phase and replaces them with the current-step gradient. This brings the optimizer closer to the pure SGD assumed in the theoretical analysis of wen2024river, at the cost of discarding useful second-order information. The empirical comparison in Exp. 2 tests whether this theoretical alignment translates into a measurable gain.